

D'Arcy Thompson and 2D mappings

Worksheet

We can represent a point, P , in the plane using a vector with components representing its x and y coordinates, i.e.

$$\mathbf{v}_P = [x_P, y_P]$$

represent a point with coordinates x_P and y_P .

A transformation is performed by defining a new point, P' , with new coordinates that are some functions of the old coordinates, i.e.

$$\mathbf{v}_{P'} = [x_{P'}, y_{P'}] = [f(x_P, y_P), g(x_P, y_P)].$$

where $f(.,.)$ and $g(.,.)$ are functions that will represent a particular transformation.

Linear transformation

Translation

Suppose that f and g are defined such that

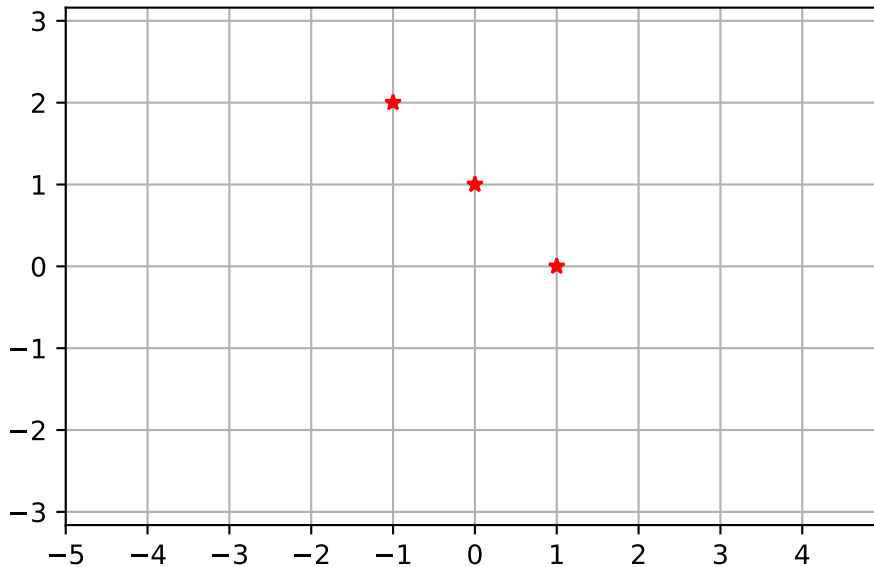
$$f(x, y) = x + t_x$$

and

$$g(x, y) = y + t_y$$

where t_x and t_y are constants.

1. Compute the transformation of the point $P(1, 2)$ in the case $(t_x, t_y) = (3, 0)$.
2. Identify the inverse of the transformation in 1.



Scalings

Consider a scaling transformation defined such that

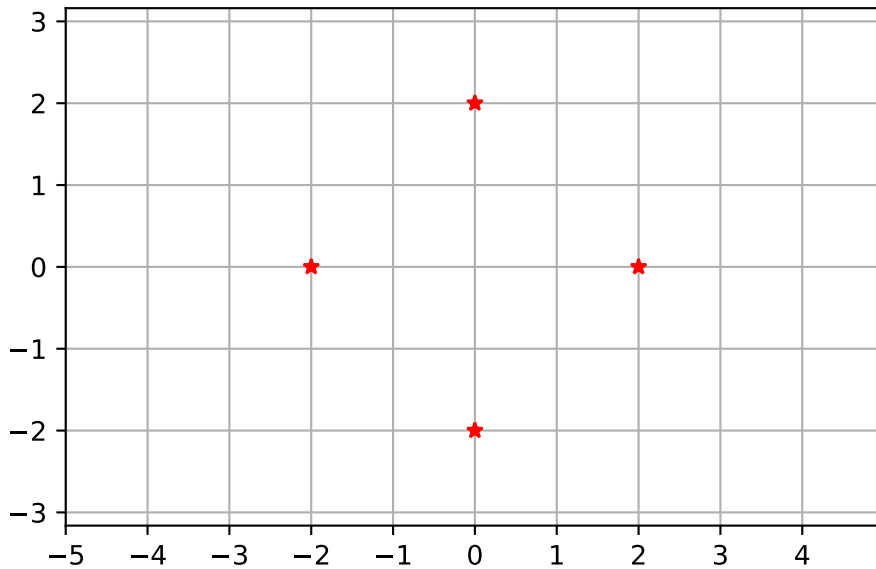
$$f(x, y) = a * x$$

and

$$g(x, y) = b * y$$

where b is a constant.

1. Compute the transformation of the point $P(2, 1)$ in the case $b = 2$.
2. Identify the inverse of the transformation in 1. Are there values of b for which an inverse does not exist?



Rotation

Now consider a transformation

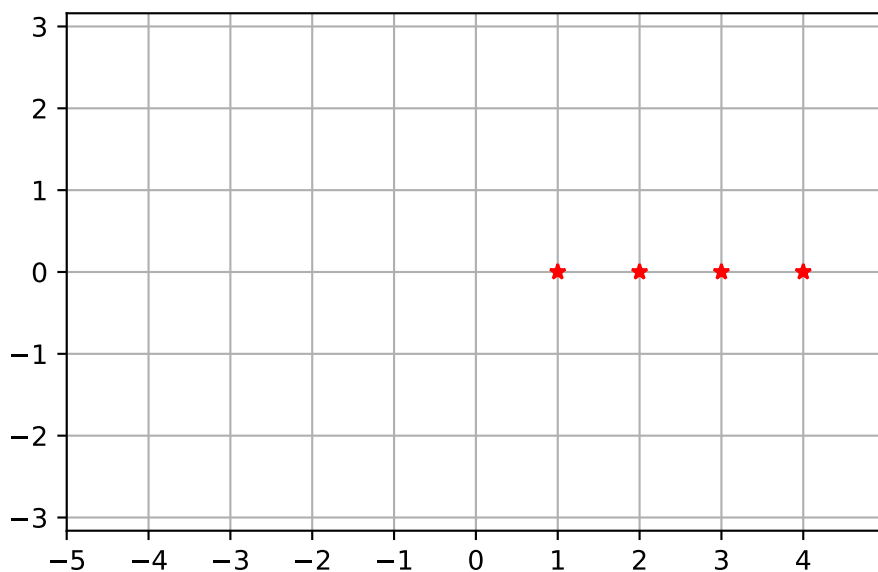
$$f(x, y) = \cos \theta x - \sin \theta y$$

and

$$g(x, y) = \sin \theta x + \cos \theta y$$

where θ is a constant value.

1. Compute the transformation of the point $P(2,0)$ in the case $\theta = 90 \deg(\frac{\pi}{2} rad)$.
2. Identify the inverse of the transformation in 1.



Shear

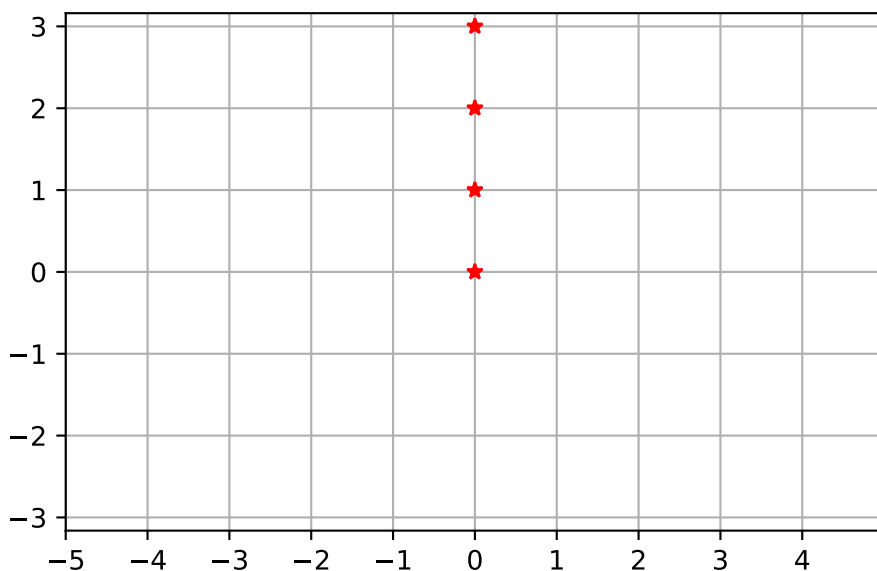
$$f(x, y) = x + ay$$

and

$$g(x, y) = y$$

where a is a positive constant.

1. Compute the transformation of the point $P(0,2)$ in the case $a = 2$.



A general representation

If you have been introduced to vectors and matrices then you may spot that the above transformations can be represented as a matrix multiplication of a vector.

Consider the matrix

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}.$$

Define

$$\mathbf{v}_{P'} = A\mathbf{v}_P$$

Can you identify values of the parameters a_{11} , a_{12} etc. that describe the rotation, shear and scaling transformations?